

Portfolio Site Report

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1.0 Introduction

The objective of this project is to create a portfolio website that serves two primary purposes.

1. Showcasing my projects – This website will provide an organised display of some of my past projects, including descriptions of my contributions, technologies used, and where applicable, interactive versions of the project for recruiters to explore.
2. Demonstrating my web development skills – Beyond acting as a portfolio, the website will demonstrate my technical capabilities in web development, allowing recruiters to assess my proficiency in building and designing user-friendly interfaces.

Additionally, I want to convey aspects of my personality and design sensibilities through a well thought out UI/UX, ensuring that the website is not just functional but also visually appealing and intuitive to navigate.

1.1 Features and Goals

- A clear, structured portfolio section with descriptions and links to projects.
- Interactive elements, allowing recruiters to engage with select projects.
- A responsive design, ensuring accessibility across various devices.
- A modern UI, reflecting my creativity and design preferences.
- Contact options, enabling recruiters to reach out easily.

1.2 Technologies

To further push my development skills, I have chosen to build the website using React.js as the foundation. While I have some experience with React, this project serves as a learning opportunity to deepen my understanding. The tech stack includes:

- Frontend: React.js, JavaScript, HTML, CSS, SASS
- Hosting & Deployment: GitHub Pages

By undertaking this project, I aim to not only present my technical skills but also challenge myself to grow as a developer by refining my knowledge of modern web technologies.

2.0 Project Planning & Initial Design

Before starting the development process, I made sketches to outline the structure of the website. I took the scrum approach when carrying out this project. I would set myself small goals, focusing on one page at a time, deciding what kinds of things I should include. I took inspiration from other websites online. I would tend to go to a website called 'Awwwards', look at some of the featured websites and try and find my own way to implement the features into my website. The focus was on ensuring easy navigation, clear project and skills sections, and a visually engaging experience.

2.1 UI/UX Considerations

The website has a minimalist design. This was chosen to maintain a clean and professional appearance that highlights projects without unnecessary distractions. The website needed to have intuitive navigation so that recruiters could quickly locate relevant sections, ensuring a seamless browsing experience. To accommodate a variety of users, a responsive layout was implemented to provide usability across desktops, tablets, and mobile devices.

3.0 Development Process

In Appendix A, images illustrating the various stages of the website's development are included. Each image will be referenced in this section to provide a clearer understanding of the build process.

3.1 Frontend Development

In the initial stage (A1), only the main page was created. Since this is a portfolio site, I prioritized designing a header with my name and desired job role in the top left corner. Below, I included indexed links to different sections of the site, even though the pages had not yet been created. A dark/light mode toggle was also added for user preference customization.

Moving forward (A2), I structured the projects, contact, and information sections. The main page displayed my contact details (LinkedIn, Email, GitHub). Clicking the 'View Arsenal' button would reveal a list of programming languages I am proficient in, replacing the contact details. The project titles were listed in a scrollable container, with interactive project images displayed to the right. Clicking an image would expand it, allowing the user to interact with the project.

A significant layout revision took place in A3. The navigation bar moved to the top right while my name remained in the top left. This fixed header ensured visibility while scrolling. The main page prominently featured my most recent university project, displaying a rotating 3D model related to the project. A button at the bottom directed users to the projects section.

Further enhancements followed (A4–A8):

- A4: A skills section was introduced, where hovering over a language highlighted my proficiency level.
- A5: The project page was refined, making the scrollable container cleaner and displaying project previews only on hover.
- A6: The information section was updated, and a footer was added to display contact details.
- A7: A dedicated project page was developed. The same layout structure was maintained for all projects, dynamically updating based on which project was selected. A Bento box was added for image display, and interactive project previews were placed at the top.
- A8: The FAQ section replaced the previous information section.

3.2 Challenges Faced & Solutions

- Challenge: Optimizing load times while using high-quality images.
 - Solution: Implemented lazy loading and compressed images for better performance.
- Challenge: Creating an interactive project showcase without compromising responsiveness.
 - Solution: Used React state management to dynamically display projects and their details.

4.0 Development Process

4.1 Key Sections

- Home Page: Displays my latest project and includes navigation options.
- About Me: Brief introduction to my skills, experience, and career aspirations.
- Projects: Detailed descriptions of my projects, including links and interactive demos.
- Contact: Various methods to reach me, including an interactive contact form.

4.2 Interactive Elements

- Project Showcases: Users can interact with projects and expand them for more details.
- Dark/Light Mode Toggle: Enhances user accessibility and personalization.

4.3 API Integrations & External Tools

- Framer Motion was utilized for smooth animations.
- EmailUS handles contact form submissions without needing a backend.
- Three.js was integrated for displaying 3D models in projects.

5.0 Iterative Improvements

5.1 Version History & Enhancements

The website underwent multiple iterations, incorporating feedback from peers and recruiters:

- Version 1.0: Basic portfolio structure with static project listings.
- Version 2.0: Implemented React interactivity, smoother animations.
- Version 3.0: Improved UI/UX, optimized performance, added contact form.
- Version 4.0: Added interactive project showcases, 3D model integration, and refined layout.

5.2 Feedback & Adjustments

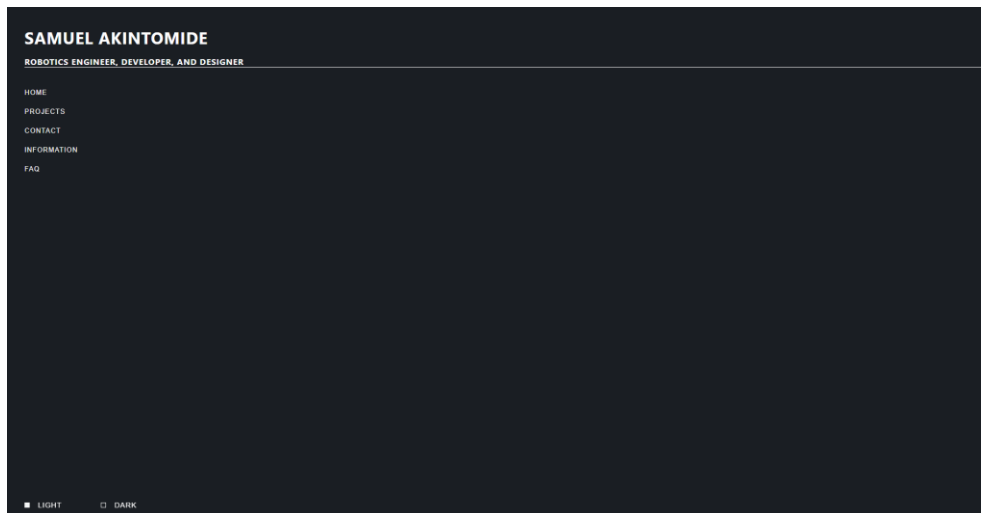
- Recruiters suggested a more intuitive navigation system, leading to the header redesign.
- Users requested project previews, resulting in the interactive scrolling project section.
- Performance concerns prompted lazy loading and code optimization.

5.3 Screenshots & Progress Tracking

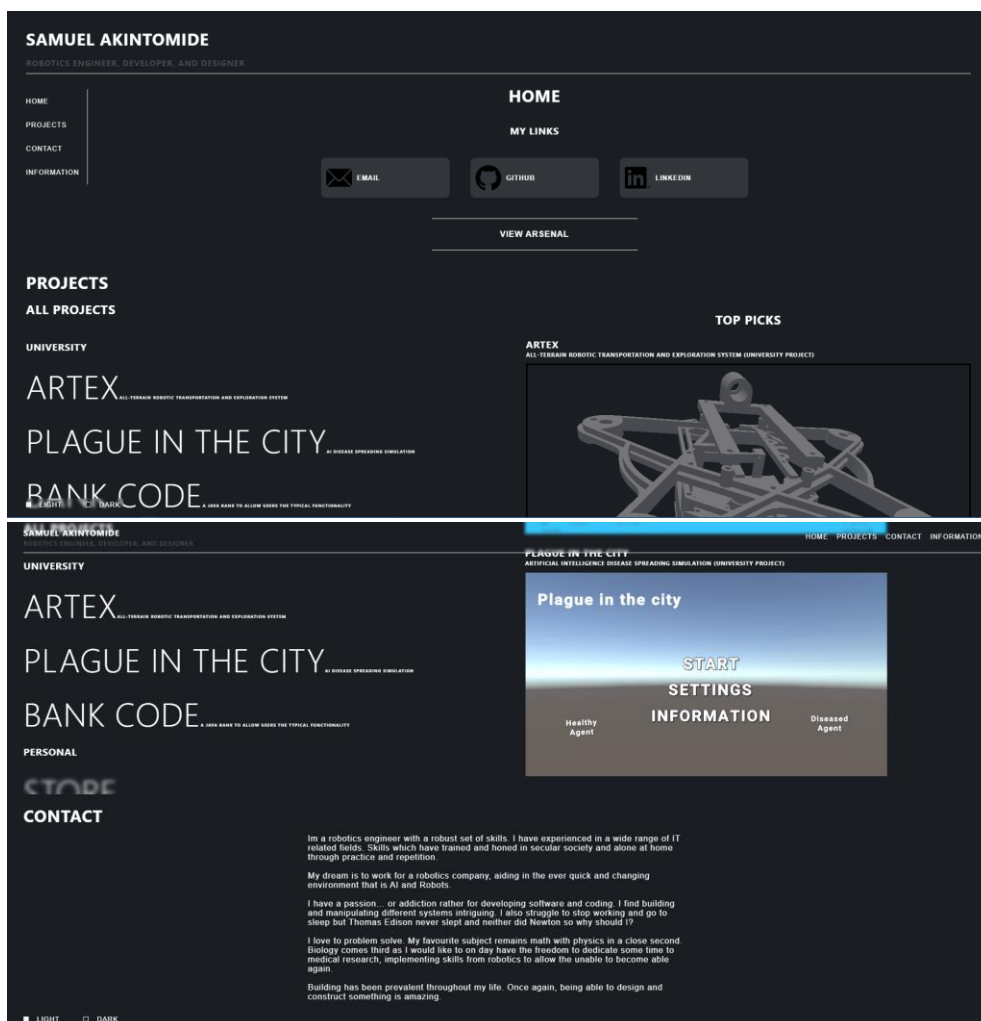
Each development phase is documented in Appendix A, displaying the transformation from the initial concept to the final product.

Appendix A

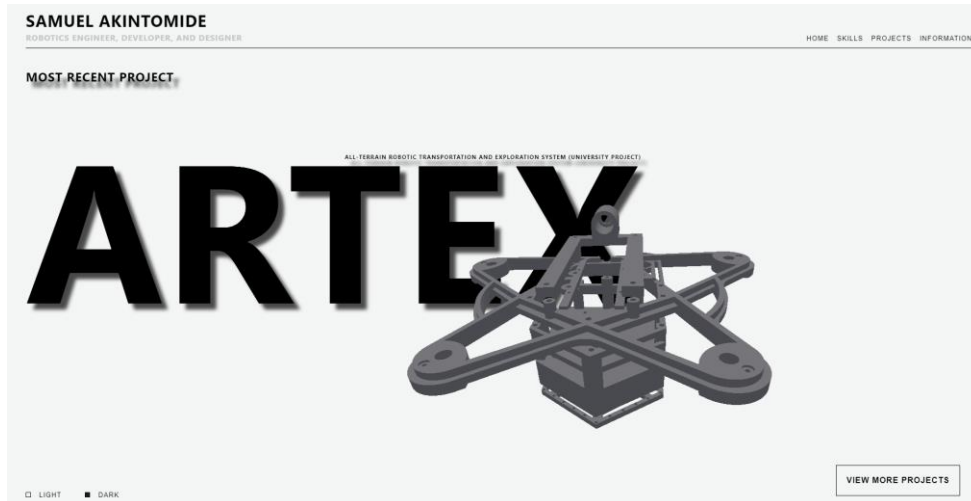
A1 – Initial Front-Page Build



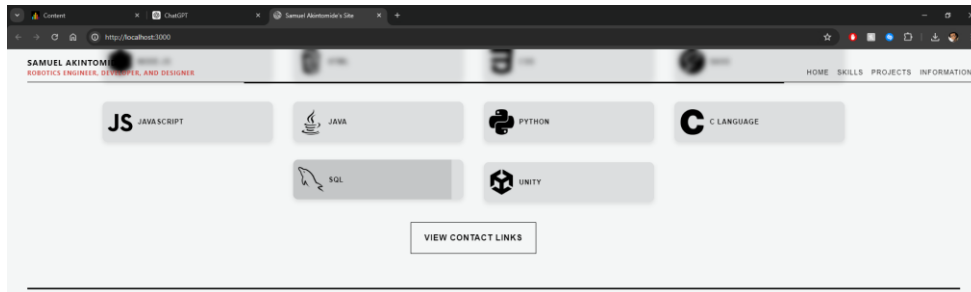
A2 – Added Project, Contact and Info Section



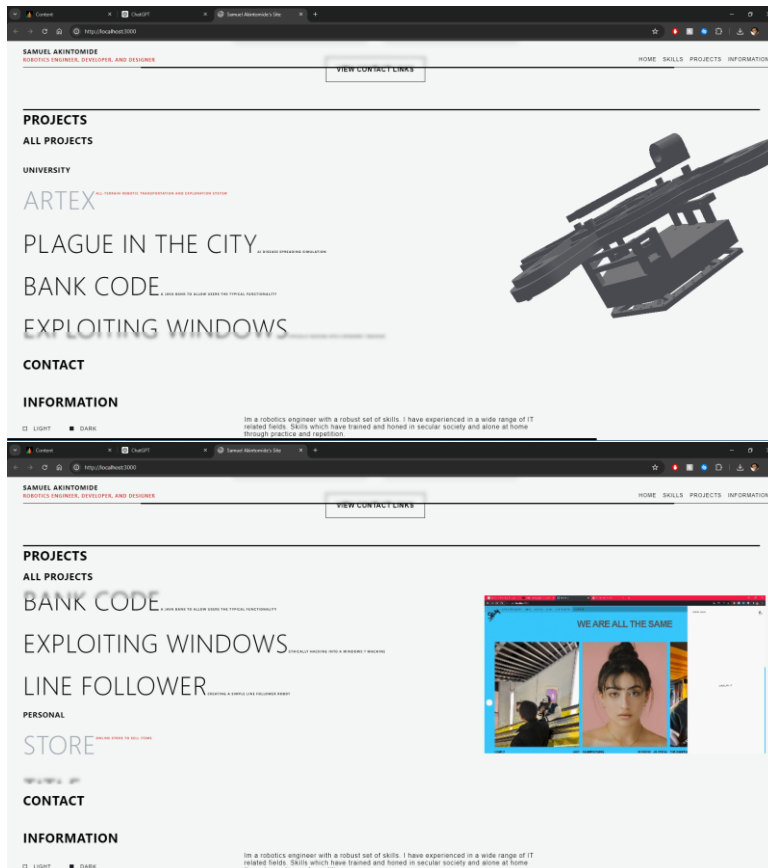
A3 – Main Page Layout Update



A4 – Skills Page Implementation



A5 – Project Page Update



A6 – Information and Contact Sections Rework



A7 – Project Page Template Build



A8 – Information Page Reworked Into FAQ

EXPLOITING WINDOWS

FAQ

Who are you?

Im a robotics engineer with a robust set of skills. I have experienced in a wide range of IT related fields. Skills which have trained and honed in secular society and alone at home through practice and repetition.

What are you aiming for?

My dream is to work for a robotics company, aiding in the ever quick and changing environment that is AI and Robots.

Do you have any passions?

I have a passion... or addiction rather for developing software and coding. I find building and manipulating different systems intriguing. I also struggle to stop working and go to sleep but Thomas Edison never slept and neither did Newton so why should I?

I love to problem solve. My favourite subject remains math with physics in a close second. Biology comes third as I would like to on day have the freedom to dedicate some time to medical research, implementing skills from robotics to allow the unable to become able again. Building has been prevalent throughout my life. Once again, being able to design and construct something is amazing.

CONTACT ME HELP HELP

LINKEDIN

EMAIL

PHONE

☐ LIGHT ☒ DARK

FOUND IN

UNITED KINGDOM

LONDON